Safe and Trustworthy
Artificial Intelligence



JUNAID QADIR

INFORMATION TECHNOLOGY UNIVERSITY
(ITU), LAHORE, PAKISTAN

3 points for this talk

(1) Safety (Is it safe to run AI/ML in mission-critical applications in the presence of adversaries/malicious actors?)

(2) Trustworthiness (Can we trust that AI/ML applications will be used for human-beneficial ends and not to harm people)

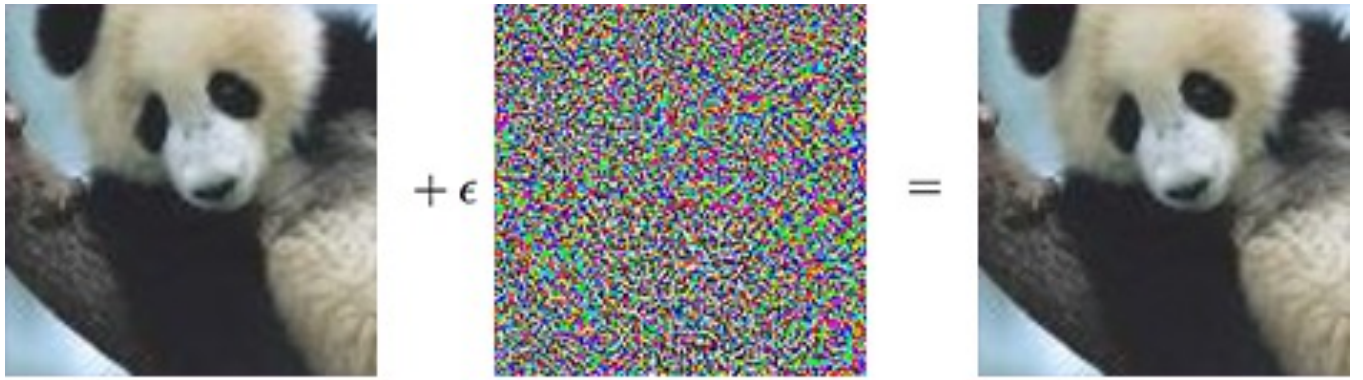
Fairness and Justice *(Are AI/ML decisions just and fair to all?)*
Bias *(Can the AI/ML model learn our biases?)*

(3) How to Design Safe & Trustworthy AI/ML systems?



Safety/Robustness of ML/AI models

Adversarial attacks on AI/ML systems





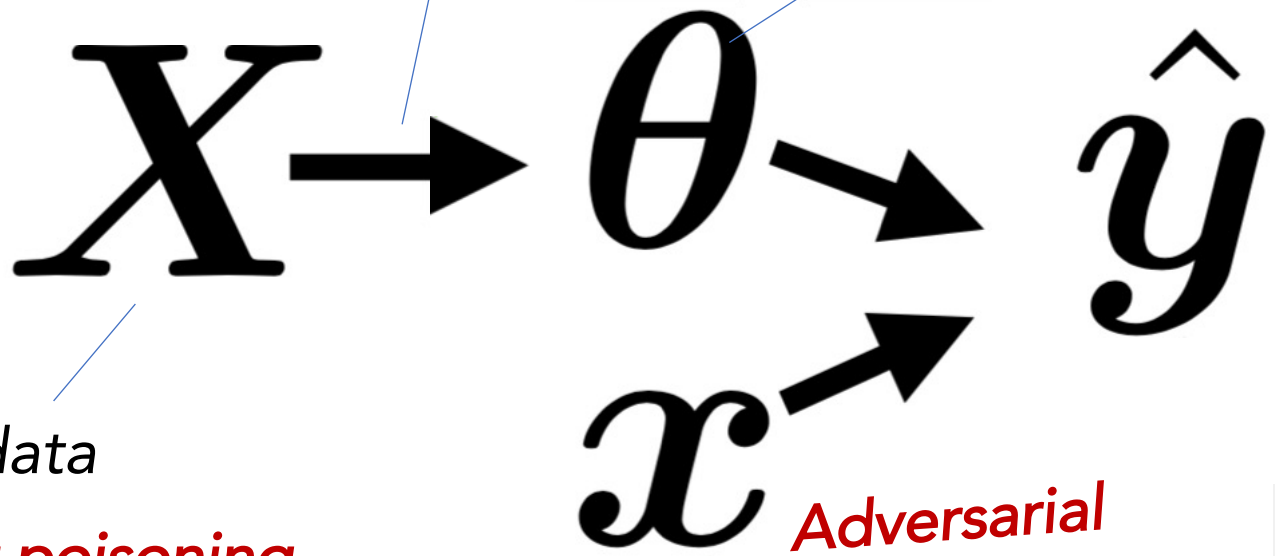
*Adversarial examples are inputs to ML models that an **attacker has intentionally designed** to cause the model to make a mistake*

All parts of the ML pipeline can be attacked

Recovery of sensitive/ private training data

Learning algorithm

Learned parameters



Test output

Model theft

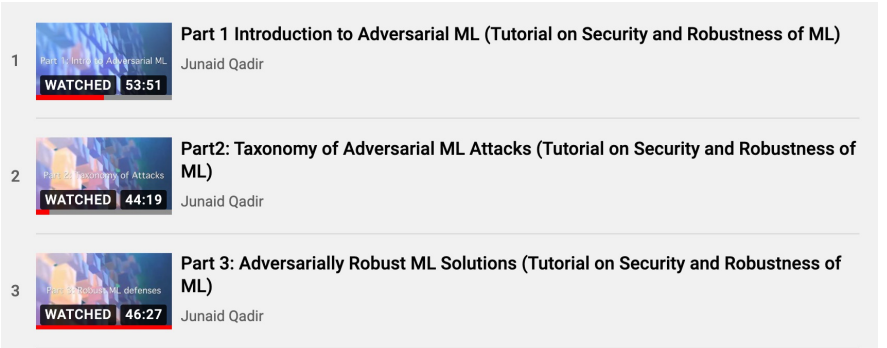
Training data

Training set poisoning

Test input

Adversarial Attack

For more:
SMARTTECH 2020 Tutorial



Available online [or on request]

Trustworthy AI/ML systems

1) *How is the technology that I am working on going to be used?*
2) *Who will benefit or suffer from it?*
3) *How much and what social impact will it have?*

Bias in training data



Image/face recognition system

PROPUBLICA



Machine Bias

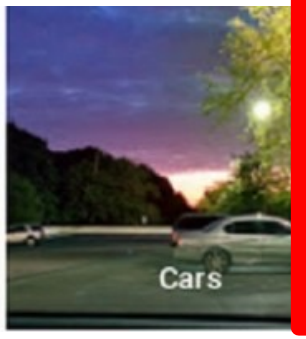
There's software used across the country to predict future criminals. And it's biased against blacks.

by Julia Angwin, Jeff Larson, Surya Mattu and Lauren Kirchner, ProPublica

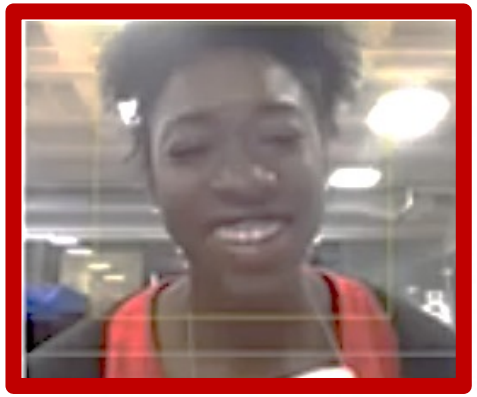
May 23, 2016

results in adverse impact









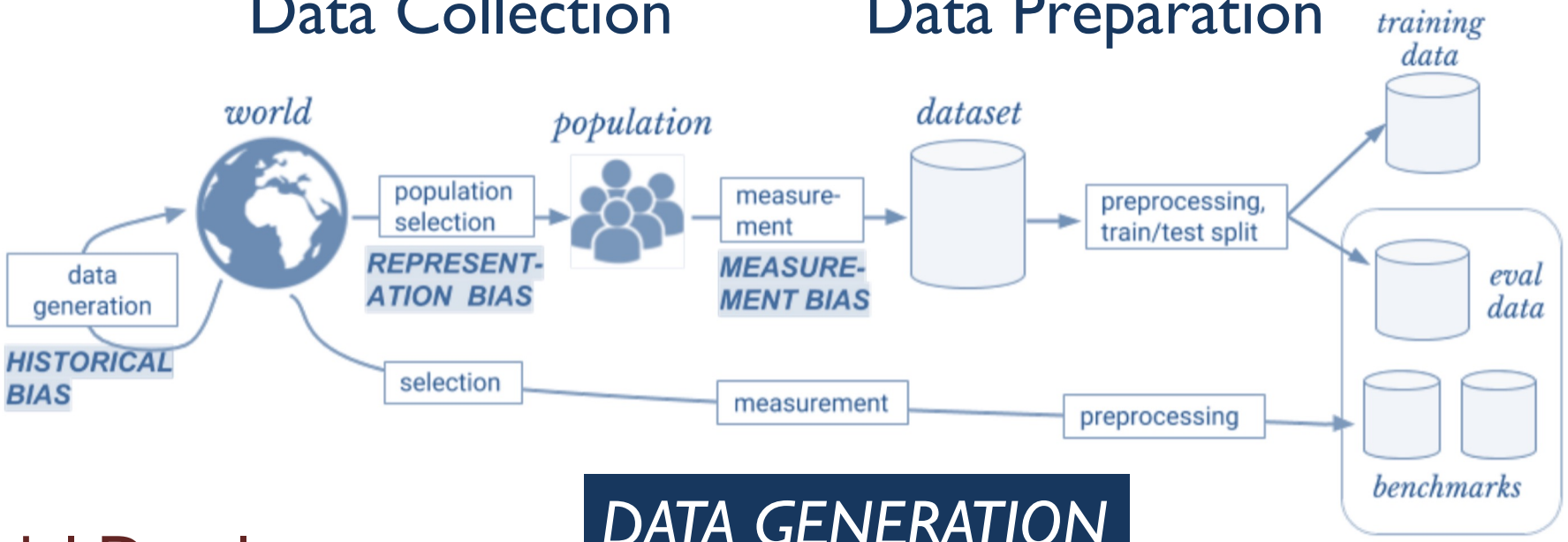


But the bias does not only come through the data

Suresh and Guttag,
A Framework for Understanding Unintended Consequences of Machine Learning
https://arxiv.org/pdf/1901.10002.pdf

Data Collection

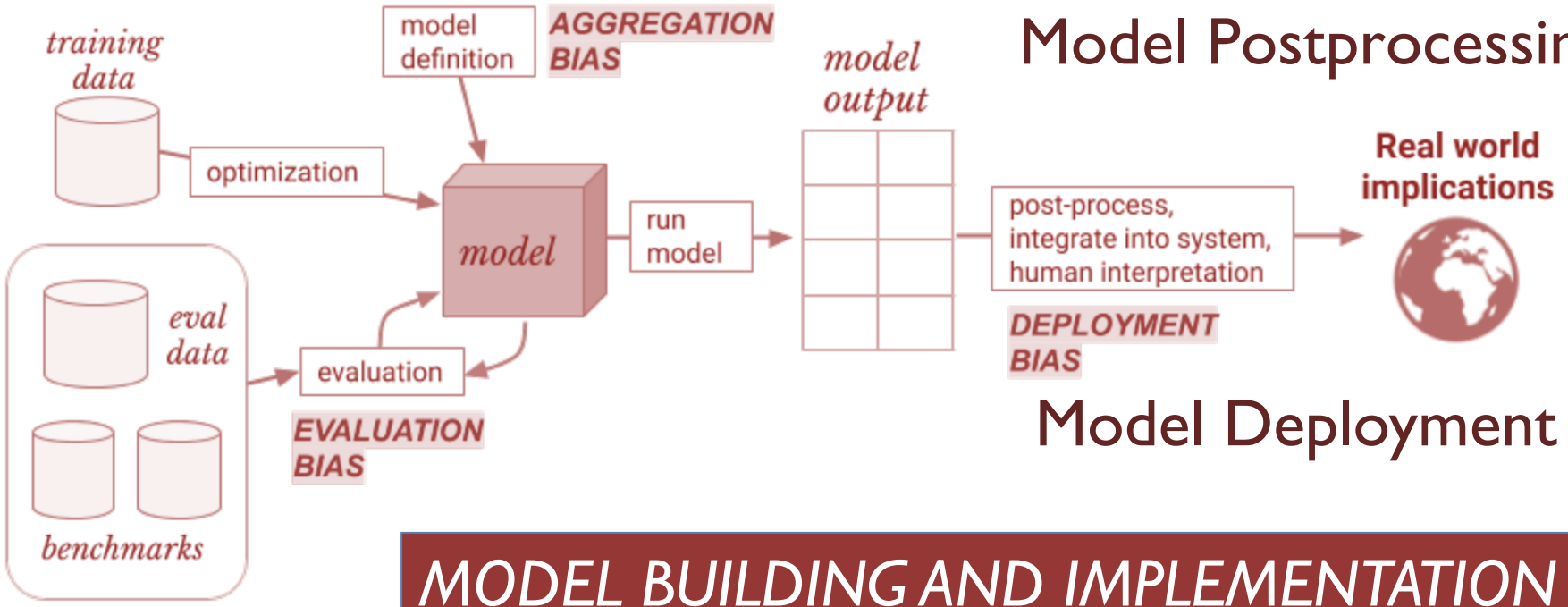
Data Preparation



DATA GENERATION

Model Development

Model Postprocessing



Model Evaluation

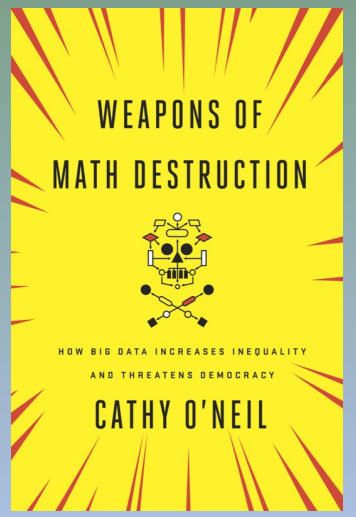
Model Deployment

MODEL BUILDING AND IMPLEMENTATION

Developing Safe & Trustworthy AI/ML systems

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THERE ARE ETHICAL CHOICES IN EVERY SINGLE ALGORITHM WE BUILD

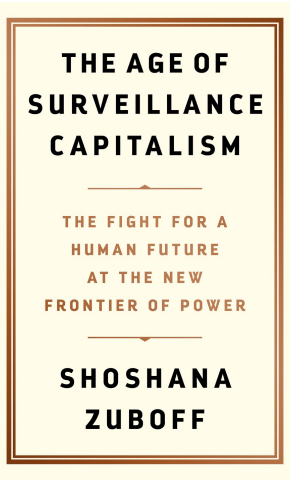


Unanticipated (possible inadvertent) harms





REWARD FUNCTION GAMING







Center for Humane Technology

Ledger of Harms

Making Sense of the World
Misinformation, conspiracy theories, and fake news

Politics and Elections
Propaganda, distorted dialogue & a disrupted democratic process

Systemic Oppression
Amplification of racism, sexism, homophobia and ableism

The Next Generations
From developmental delays to suicide, children face a host of physical, mental and social challenges

Social Relationships
Less empathy, more confusion and misinterpretation

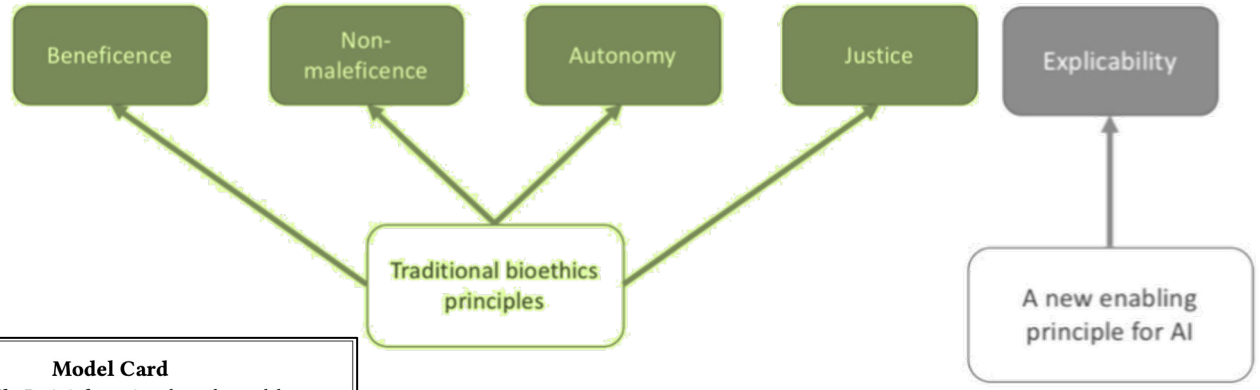
Physical and Mental Health
Stress, loneliness, feelings of addiction, and increased risky health behavior

nature machine intelligence
PERSPECTIVE
https://doi.org/10.1038/s42256-019-0088-2

The global landscape of AI ethics guidelines

Anna Jobin, Marcello Ienca and Effy Vayena*

The four high-level ethical principles capture the essence of most AI declarations



nature machine intelligence
PERSPECTIVE
https://doi.org/10.1038/s42256-019-0114-4

Principles alone cannot guarantee ethical AI

Brent Mittelstadt



This calls for a multipronged approach combining public policy, regulation, economic incentives, accountability as well as technical research.

Model Card

- **Model Details**. Basic information about the model.
  - Person or organization developing model
  - Model date
  - Model version
  - Model type
  - Information about training algorithms, parameters, fairness constraints or other applied approaches, and features
  - Paper or other resource for more information
  - Citation details
  - License
  - Where to send questions or comments about the model
- **Intended Use**. Use cases that were envisioned during development.
  - Primary intended uses
  - Primary intended users
  - Out-of-scope use cases
- **Factors**. Factors could include demographic or phenotypic groups, environmental conditions, technical attributes, or others listed in Section 4.3.
  - Relevant factors
  - Evaluation factors
- **Metrics**. Metrics should be chosen to reflect potential real-world impacts of the model.
  - Model performance measures
  - Decision thresholds
  - Variation approaches
- **Evaluation Data**. Details on the dataset(s) used for the quantitative analyses in the card.
  - Datasets
  - Motivation
  - Preprocessing
- **Training Data**. May not be possible to provide in practice. When possible, this section should mirror Evaluation Data. If such detail is not possible, minimal allowable information should be provided here, such as details of the distribution over various factors in the training datasets.
- **Quantitative Analyses**
  - Unitary results
  - Intersectional results
- **Ethical Considerations**
- **Caveats and Recommendations**

Quantitative Analyses



Conclusion

1. Ethical AI Design requires the confluence of many fields (technical, ethical, philosophical, legal, political)

2. AI has the potential for huge benefit and massive harm (individual as well as social). There is an urgent need to focus not only on efficient AI/ML but also on safe and trustworthy AI/ML

Questions/Comments?

Junaid.qadir@itu.edu.pk